I am a third-year Computer Engineering student at the **University of Bologna**, currently on exchange at **Yonsei University** in Seoul. My interest lies in **computing hardware and systems**, and in how they can connect with other disciplines.

I have already had a first taste of research. Last summer I completed a research internship in **Ben Engel's lab at the Biozentrum** in Basel, where I worked on several projects: mainly **expansion microscopy**, but also **motif search in protein sequences** and **tomography reconstruction with AI**. I took care of implementing a **Python library** which used a **pre-trained AI model to interpolate tomograms** in the pipeline of the group, I expanded and imaged microscopic organisms (*Chlamydomonas* and *Chlorella* mainly) and used motif search tools to try to replicate an observed pattern from many proteins. I developed practical experience with the lab instrumentations and I understood how research actually works, and I am now sure I want to continue in this direction, ideally with a **PhD**.

I like to bring computing into other fields. I don't think I want to do research in pure computer science; I would rather apply my skills somewhere else. I am particularly interested in **space and astronomy**, but I also love technology inspired by nature, and I want to learn more about **brain simulation through circuits**.

My background is mainly in **low-level programming**. I have been studying operating systems and programming since high school, partly through self-study: I worked through **OSTEP** and followed **Nand2Tetris**, which gave me a first understanding of computer architecture. At Yonsei I am taking **Digital Microelectronics**, **Electronic Circuits**, **Deep Neural Networks**, and **System Programming**, among others. At TU/e Eindhoven next semester I will add **Machine Learning**, **Signals and Systems**, and **Control Systems**. For my bachelor thesis I plan to work with the **PULP group** in Bologna on **hardware AI accelerators** or **fault-tolerant design**. I have also been part of **student engineering teams** at Bologna while keeping a **high GPA**.

My three choices, in order:

Integrated Neurotechnology Laboratory (Prof. Shoaran). The point where electronics meets the brain is the area I find most compelling. A workshop by Professor Flavio Donato at the Biozentrum on how brain activity is organized and activated fascinated me. I looked into the group's recent research, and among the papers I read, **LogSync** caught my attention. I would like to work on the **architecture design of chips** like this one, integrating **signal and data analysis with machine learning**. I believe I could contribute thanks to my knowledge of **ML, microelectronics, and firmware**.

Biomedical and Neuromorphic Microelectronic Systems group (Prof. Schmid). I am drawn to this group because it takes the brain, and nature more broadly, as a **design principle for hardware**. I went through Prof. Schmid's last papers and I spent most of my time on **neuromorphic computation-in-memory** with **binary neural networks** on **magnetic tunnel junctions**. I had already heard of **in-memory computing** from a research group in Bologna so that's why it caught my eye. I am also interested in the group's work on **reservoir computing** and I would be curious to see how it could meet in-memory computation. I would like to contribute to the **design or testing** of such systems, depending on the group's current projects.

Embedded Systems Laboratory (Prof. Atienza). I enjoy **firmware and low-level development**; it is the sub-field of computer science I like most. I am particularly interested in **power consumption**, which I expect to become a central constraint as AI workloads grow: I recently learned about the Linux **Energy Aware Scheduler (EAS)**, which sparked my curiosity about **energy-efficient computing**. Looking at the lab's research, I found **X-HEEP** especially interesting: I am a strong supporter of **open source**, and it is an open **RISC-V** platform I would like to contribute to. Through my interest in PULP I have become drawn to **fault-tolerant architectures**, which fit my passion for aerospace, and I would gladly contribute to **space-oriented work**: I saw the lab's earlier project on **embedded AI for aerospatial navigation**, and I would be glad to join similar work if the lab is pursuing it.

For me this program is above all a chance to learn from people doing research and to grow as a researcher.

Sincerely,
Simone Rea